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Climate policy uncertainty and labor demand in high-carbon industries

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ABSTRACT

Using data from Chinese high-carbon companies, this paper documents a negative association between climate policy uncertainty and corporate labor demand. This finding remains valid after implementing a series of robust and endogeneity tests. The research results also indicate for firms in poorer economic situation regions, firms with fewer government subsidies, more high tech employees and lower profitability, the negative effect of climate policy uncertainty on corporate labor demand is greater. The mechanism indicates that climate policy uncertainty affects corporate labor demand through green investment and labor outsourcing.

1. Introduction

Climate risks caused by extreme weather and climate events have decelerated economic growth (Dell et al., 2008; Dellink et al., 2014) and posed challenges financial stability (Campiglio et al., 2018; Battiston et al., 2021). Governments frequently implement climate policies aimed at mitigating the impacts of climate risks (Gavriilidis, 2021; Battiston et al., 2021). In turn, numerous controversies and disagreements over various climate policies may lead to a series of adverse economic consequences, such as diminishing firms' productivity (Ren et al., 2020; Dai and Zhu, 2024), decreasing energy investing, but increasing demand for renewable energy (Bogmans et al., 2024; Shang et al., 2022), altering banks' risk (Dai and Zhang, 2023; Liu et al., 2024a), and enhancing corporate green innovation (Bogmans et al., 2024; Gavriilidis, 2021; Bai et al., 2023). However, much less is known about how firms adjust labor investing decisions in response to perceived increases climate policy uncertainty (CPU).

Human capital is a core competitive force for long-term growth at the firm level (Barney, 1991; Delery and Roumpi, 2017), influencing both corporate innovation (Østergaard et al., 2011; Chemmanur et al., 2019) and performance (Crook et al., 2011; Buller and McEvoy, 2012). In this regard, the extant literature have explored the effects of specific environmental policies on labor demand, such as green finance policy (Jiang and Jiang, 2023; Xie et al., 2024) and carbon emissions governance (Ren et al., 2022a; Wang et al., 2023). Notably, economic policy uncertainty is negatively associated with labor demand Nguyen and Phan (2020) and Chu and Fang (2021). We take a step further to contend that firms' labor demand, an important investment decision, could be influenced by uncertainty arising from climate policy within the high-carbon industry setting in China.[1]

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[1] We define high-carbon industries followed by Wang et al. (2024), include electric power, heat, gas, petroleum, non-ferrous and ferrous metal smelting and rolling processing industry, chemical industry, construction, paper making industry, machinery and other related industries.

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Climate and employment issues have become central to the United Nations Sustainable Development Goals (Fuso Nerini et al., 2019; Halkos and Gkampoura, 2021). Clarifying the relationship between CPU and labor demand can help countries balance employment and climate policies to promote sustainable development. China, as the world's largest carbon emitter, still relies on fossil energy (Iqbal et al., 2024), and its climate policies have important global implications (Cai et al., 2021; Iqbal et al., 2024). China also has the largest labor force in the world (779 million people, World Bank, 2023), therefore, studying China is exemplary. In addition, firms are under pressure of low-carbon transition, with high-carbon emission firms taking responsibility for reducing emissions as major emitters (Liu et al., 2024b). It is representative to study the transition challenges of high-carbon emission industries.

In this paper, we propose that CPU has a negative effect on corporate labor demand. We present two major reasons for this relationship. First, based on the wait-option theory, we contend that as policy uncertainty increases the value of the wait option (Bloom, 2009; Miller and Park, 2002), firms tend to reduce or postpone investment as a response until the uncertainty is resolved (Chen et al., 2019; Wen et al., 2022). This argument is consistent with Greenland et al. (2019), Dixit et al. (1994), and Atsu and Adams (2021), who argue that firms are likely to delay decisions involving significant costs if those decisions are difficult to reverse under uncertainty. It can be inferred that corporate labor demand (regular employees), a type of investment, is also reduced in such challenge times.

Second, in light of the financial friction theory, a rise in CPU exerts broader financial constraints on firms due to the increased information asymmetry between firms and outside investors, as evidenced in the literature (Gilchrist et al., 2014; Ren et al., 2022b). Several scholars provide empirical evidence that, when facing policy uncertainty, creditors tend to raise financing costs, which increases cash flow volatility for firms and contributes to greater financing constraints and delays in labor investment decisions (Pástor and Veronesi, 2013; Tran, 2021; Li et al., 2024). Due to the lack of sufficient funds, firms may struggle to hire more workers, ultimately resulting in lower labor demand (Benmelech et al., 2021; Mbanyele et al., 2024). Collectively, we predict that firms are likely to adopt a more conservative labor decisions as a function of the increase in CPU.

The results reveal a statistically significant negative link between CPU and corporate labor demand in high-carbon industry and support the wait-option and the financial friction theory. Our findings remain valid even after applying several robust and endogeneity tests that include a quasi-natural experiment, and Instrumental Variable test. The heterogeneity tests show that the negative impact of CPU on firms' labor demand is greater for firms in poorer economic situation regions, firms with fewer government subsidies, more high tech employees and lower profitability. The mechanistic analysis suggest that CPU affects firms' labor demand through green investment and labor outsourcing.

This paper contributes to the literature in three major ways. Firstly, enhancing the literature on the economic effects of CPU and related policy factors associated with labor demand. Existing literature mainly focuses on the influence of CPU on the energy market (Bogmans et al., 2024; Shang et al., 2022), stock market dynamics (Bouri et al., 2022; Xu et al., 2023), and productivity (Dai and Zhu, 2024; Ren et al., 2022b). And green financial policy (Jiang and Jiang, 2023; Xie et al., 2024), economic policy uncertainty (Li et al., 2025; Nguyen and Phan, 2020) and low carbon governance (Wu et al., 2024; Wang et al., 2024) are the main policy factors that influence labor demand. To our knowledge, this paper is the first to examine the impact of CPU on corporate labor demand.

Secondly, this paper finds mechanisms of how CPU affects labor, providing new support for the application of waiting option theory and financial friction theor. In particular, in the labor segment, following Li et al. (2025) demonstration that CPU induces firms to increase labor outsourcing, this paper extends its economic consequences. Specifically, CPU boosts labor outsourcing, which has a crowding-out effect on firms' own demand for employees. This finding provides a new perspective for understanding the role of CPU on employment patterns.

Thirdly, most of the previous studies have directly employed the U.S. climate policy index to analyze the Chinese economic issues. Although the U.S. CPU has a certain impact on the world, it is difficult to directly affect Chinese firms (Mo and Liu, 2023; Iqbal et al., 2024). This paper introduces the newly constructed Chinese CPU index by Ma et al. (2023) for empirical evidence, which contributes to testing the applicability of this index in Chinese labor research.

2. Data and methods

2.1. Data

Our empirical analysis employs a sample of listed firms in Chinese high-carbon industries from 2007 to 2022,[2] excluding ST, *ST, and PT samples. All continuous variables are winsorized at the 1st and 99th percentiles to mitigate the influence of extreme values. Firm level and macroeconomic data are sourced from the CSMAR database and the National Bureau of Statistics of China respectively. The final sample comprises 9978 firm-year observations.

[2] The Chinese government implemented new accounting rules in 2007.

2.2. Variables and model specification

We define the natural logarithm of total employees of a given firm to measure labor demand (Labor) of each firm, which has been used by Ren et al. (2022a). Chinese CPU index are extracted from Ma et al. (2023).[3] This index is derived from more 1.7 million articles from six mainstream Chinese newspapers. The higher the index, the larger climate policy uncertainty.

Following Jiang and Jiang (2023), we control for a vector of variables that may shape labor demand. These variables include firm age (Age), firm size ($Size$), leverage ratio ($Leverage$), sales growth ($Growth$), Tobin Q ($Tobinq$), operating cash flow ($Cashflow$), board size ($Board$), percentage of independent directorso ($Independent$), Duality ($Dual$) and percentage of largest shareholding ($Top1$). To ease the potential bias from omitting time effects, following Li et al. (2025) and Kugler and Saint Paul (2000), we further incorporated China's economic policy uncertainty (Epu),[4] unemployed population ($Unemployment$) and growth of total national GDP (Gdp_growth) as the macroeconomic control variables.

The definitions of each variable are presented in Table A.1. We construct Eq. (1) to further verify the impact of CPU on labor demand in the Chinese high-carbon industry.

$$Labor_{i,t+1} = \beta_0 + \beta_1 Ccpu_t + \beta_2 ControlVariables_{i,t} + \delta_{ind} + \gamma_{city} + \theta_i + \varepsilon \tag{1}$$

where i indexes the firm and t is the year. We use corporate labor demand in the following year as the dependent variable and regress it against $Ccpu$. δ_{ind}, γ_{city} and θ_i indicates industry, city and firm fixed effects as well as clustering standard errors at firm level. Year fixed effects are not introduced, because there is no difference in $Ccpu$ corresponding to all firms in the same year.

2.3. Quasi-natural experiment

The signing of the Paris Agreement provides an excellent setting for us to identify how firms in high-carbon industries adjust their labor demand. Empirically, we assign the dummy variable '$Posti$' a value of 1 if the year falls within the 2016 to 2022 period, and 0 otherwise. Meanwhile, losing firms tend to be more sensitive to macro policies, as firms tend to attribute their losses entirely to the burdens imposed by macro policies and seek policy supports after loss had been recognized (Lin and Tan, 1999; Liao et al., 2009). Nonetheless, policy uncertainty increases the difficulty of securing policy supports. Therefore, Firms have loss are used as the treatment group ($Treat_{i,t} = 1$), while the remaining firms are placed in the control group ($Treat_{i,t} = 0$).

We use the Differences-in-Differences method (DID) and construct the following model to verify this path:

$$Labor_{i,t+1} = \beta_0 + \beta_1 Post_i * Treat_{i,t} + \beta_2 Post_i + \beta_3 ControlVariables_{i,t} + \delta_{ind} + \gamma_{city} + \theta_i + \varepsilon \tag{2}$$

where i indexes the firm and t is the year. δ_{ind}, γ_{city} and θ_i indicate industry, city and firm fixed effects as well as clustering standard errors at firm level. Year fixed effects are not introduced as we control the model using a number of macro variables (no difference in the same year) and $Post_i$.

2.4. Descriptive statistics of main variables

Table 1 presents descriptive statistics for main variables. The mean and standard deviation of $Labor_{t+1}$ are 7.9227 and 1.2314, respectively, aligning with findings from Ren et al. (2020). For the China's climate policy uncertainty ($Ccpu$) index, we observe a mean of 2.4428 and a standard deviation of 0.4593, consistent with Ma et al. (2023).

3. Empirical results

3.1. Baseline regression

Table 2 reports the results of the baseline regressions on the effect of CPU on labor demand with different control conditions. The results show that the regression coefficients of CPU are all significant and negative, indicating that CPU has a significant negative impact on the labor demand of high-carbon emission firms. This is consistent with the theoretical expectation in the previous section, and a reasonable explanation is that CPU raises the value of waiting options as well as financial frictions, which leads firms to reduce their investment in labor.

3.2. Quasi-natural experiment and parallel trend test results

Consequently, Column (1) of Table 3 reports the regression results of model (2). The coefficient of $Posti*Treat_{i,t}$ is significantly negative, support the above argument. Fig. 1 represents the results of the parallel trend test. Before $Current$, the coefficients are not significant. However, the coefficients tend to be significantly negative from the second year after $Current$, indicating the continued effect of Paris Agreement. Clearly China's climate policies are being released more frequently after signing the Paris Agreement. It has a greater impact on the labor demand of high-carbon emission firms, especially the losing firms. This also supports the results of our baseline regression that CPU has a negative effect on the labor demand of high-carbon emission firms.

[3] China's annual climate policy uncertainty index represented on https://figshare.com/articles/dataset/China_s_CPU_index/24071193?file=45565755.

[4] China's economic policy uncertainty index represented on https://www.policyuncertainty.com/scmp_monthly.html.

Table 1
Descriptive statistics.

Variable	Observation	Mean	Std. dev.	Min	Max
Labor	9978	7.9227	1.2314	4.4998	11.2260
Ccpu	9978	2.4428	0.4593	1.3632	3.1998
Age	9978	2.3444	0.6655	0.6931	3.4012
Size	9978	22.2797	1.3386	19.8122	26.3017
Leverage	9978	0.4222	0.1975	0.0563	0.8919
Growth	9978	0.1724	0.3675	−0.5724	2.4993
TobinQ	9978	2.0430	1.3167	0.8522	8.5446
Cashflow	9978	0.0603	0.0673	−0.1594	0.2467
Board	9978	2.1670	0.2076	1.0986	2.8904
Independent	9978	0.3704	0.0512	0.3000	0.5714
Dual	9978	0.2275	0.4192	0	1
Top1	9978	0.3609	0.1516	0.0848	0.7430
IV_Cpu	9978	1.3974	0.4942	0.7757	2.1863
Loss	9978	0.0899	0.2860	0	1
Epu	9978	3.5722	2.3787	0.8225	7.9187
Unemployment	9978	12.5524	0.5915	10.7721	13.6235
Gdp_growth	9978	0.0748	0.0240	0.0235	0.1420
Giv	8284	0.1515	0.1001	0.0000	0.5865
Outsourcing	9978	0.0372	0.1327	0.0000	0.7736
DM_ed	9978	0.4946	0.5000	0	1
DM_subsidy	9978	0.2119	0.4087	0	1
DM_tech	9978	0.1850	0.3883	0	1
DM_profitability	9978	0.5001	0.5000	0	1

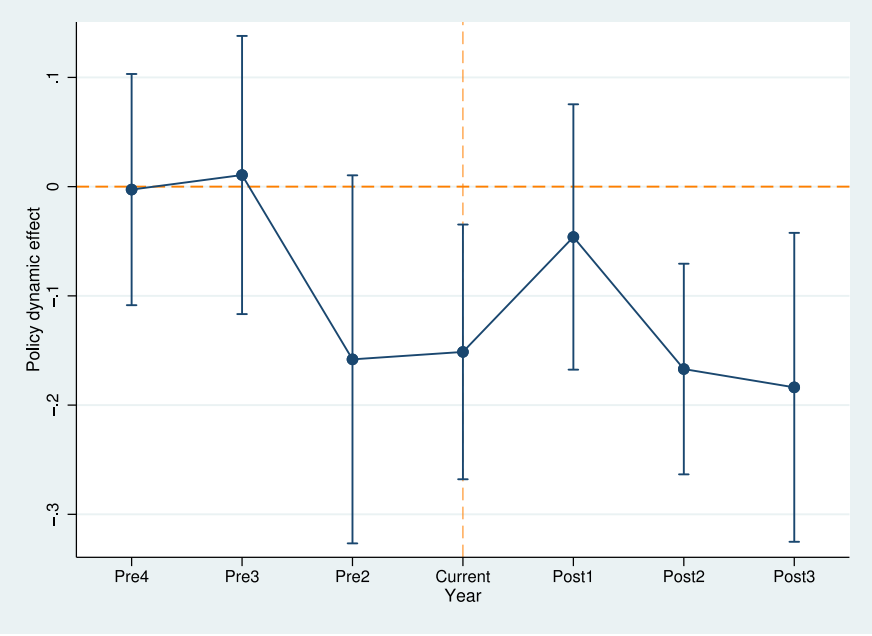


Fig. 1. Parallel trend test results.

3.3. Robust tests

3.3.1. Alternative sample test

Following Ren et al. (2022a), to minimize the effects of the global financial crisis and global pandemic, we regress baseline model after excluding observations from 2008 to 2010 and 2020 to 2022. In Column (2) of Table 3, the coefficient of $Ccpu$ is significantly negative, consistent with the baseline regression results. It suggests that the significant negative relationship between CPU and labor demand did not arise as a result of the global financial crisis and global pandemic, but existed before such shock. It further validates that CPU suppresses labor demand in high-carbon emission firms, supporting our hypothesis.

Table 2
Baseline regression results.

Variables	(1) $Labor_{t+1}$	(2) $Labor_{t+1}$	(3) $Labor_{t+1}$
Ccpu	−0.1299***	−0.2632***	−0.1636***
	(0.0268)	(0.0211)	(0.0277)
Age		0.0192	0.0832*
		(0.0129)	(0.0462)
Size		0.6946***	0.5807***
		(0.0078)	(0.0254)
Leverage		0.1301***	0.0839
		(0.0463)	(0.0808)
Growth		−0.0432**	0.0203
		(0.0208)	(0.0135)
TobinQ		−0.0047	0.0203***
		(0.0063)	(0.0063)
Cashflow		1.0226***	0.2074**
		(0.1171)	(0.0840)
Board		0.1587***	0.0960
		(0.0443)	(0.0709)
Independent		0.0745	−0.0311
		(0.1665)	(0.1837)
Dual		−0.0293	0.0365*
		(0.0187)	(0.0194)
Top1		0.4572***	0.4358***
		(0.0536)	(0.1550)
Unemployment		0.0931***	−0.0604***
		(0.0129)	(0.0197)
EPU		−0.0203***	−0.0142***
		(0.0052)	(0.0036)
GDP_growth		2.8946***	1.7975***
		(0.4606)	(0.4695)
Constant	8.2400***	−8.8972***	−4.5839***
	(0.0666)	(0.2564)	(0.5615)
Observations	9978	9978	9978
R-squared	0.0023	0.6362	0.9304
Industry FE	No	No	Yes
City FE	No	No	Yes
Firm FE	No	No	Yes
Cluster	No	No	Firm

Note: *, **, *** represents significance at 10%, 5%, and 1%, respectively. Standard errors are clustered at the firm level and reported in parentheses.

3.3.2. Instrumental variable approach

Following Mo and Liu (2023), we utilize the U.S. CPU index (IV_Cpu), provided by Gavriilidis (2021),[5] as an instrumental variable for Chinese CPU to address potential endogeneity concerns. As the largest economy, U.S. climate policy is likely to affect Chinese government policy. However, U.S. CPU is hardly likely to directly affect Chinese firms' labor decisions. Specifically, the first-stage regression confirms that the U.S. CPU is significantly positively associated with Chinese CPU, thereby validating the relevance criterion. In the second stage, we document a significantly negative relationship between predicted Chinese CPU and labor demand in high-carbon industry, which enhances the earlier conclusion. Column (3) and (4) of Table 3 report the results of two-stage least squares estimation, which not only confirm the inhibition effect of CPU on labor demand in China's high-carbon emission firms, but also demonstrate the transmission effect of CPUs across majority economies.

3.3.3. Dynamic effect

In order to test the long-term effect of CPU on the labor demand in high-carbon industries, we employ $Labor_{t+2}$ and $Labor_{t+3}$ as dependent variables for the dynamic effect test. Table 3 Column (5) and (6) indicate that the regression coefficients of $Ccpu$ on both $Labor_{t+2}$ and $Labor_{t+3}$ are negative and significant at the 1% level. These results clearly investigate a negative long-term negative effect of CPU on labor demand in China's high-carbon industries. In other words, firms may continue to make prudent long-term decision adjustments in response to CPU, and supporting our hypothesis.

[5] The U.S. Climate Policy Uncertainty index reported on https://www.policyuncertainty.com/climate_uncertainty.html.

Table 3
Robust test.

Variables	(1) Quasi-natural experiment	(2) Alternative sample test	(3) IV-First stage	(4) IV-Second stage	(5) Dynamic effect test	
	$Labor_{t+1}$	$Labor_{t+1}$	$Ccpu$	$Labor_{t+1}$	$Labor_{t+2}$	$Labor_{t+3}$
Ccpu		−0.2113***		−1.194***	−0.1283***	−0.1414***
		(0.0350)		(0.2969)	(0.0266)	(0.0253)
*Post*Treat*	−0.0735***					
	(0.0246)					
Post	−0.0880***					
	(0.0162)					
IV_Cpu			0.0735***			
			(0.0114)			
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	9978	6711	9978	9978	8419	7122
R-squared	0.9297	0.9402	0.7002		0.9329	0.9362
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Cluster	Firm	Firm	Firm	Firm	Firm	Firm
Anderson–Rubin Wald test			20.41***			
Stock-Wright LM S statistic			30.20***			

Note: *, **, *** represents significance at 10%, 5%, and 1%, respectively. Standard errors are clustered at the firm level and reported in parentheses.

4. Heterogeneity analyses

4.1. Regional economic situation

The higher level of regional economic development increases corporate labor demand (Li et al., 2024). Specifically, developed regional economics mitigate the risk of business failure through advanced policies (He and Yang, 2016), and build higher levels of economic agglomeration to allows firms to secure more financial support for expansion (Pan and Yang, 2019). All these contribute to a significant increase in labor demand in developed economically regions (Bin et al., 2020). Following Li et al. (2024), this paper employs the per capita GDP of the given year to measure the level of regional economic development (Ed). DM_ed is a dummy variable which equals 1 if Ed is greater than the annual median value and 0 otherwise. Columns (1) of Table 4 suggests that the negative effect of CPU on labor demand is pronounced for firms in regions with worsen economic conditions.

4.2. Government subsidy

Government subsidies not only help alleviate firms' labor costs, but also highlight the value of firms from helping them better cope with the CPU (Li et al., 2025). Thus, firms that receive government subsidies are usually better able to cope with this uncertainty, implying that CPU has a weaker inhibition effect on the labor demand of these firms. We define $DM_subsidy$ to 0 and 1 based on the annual city industry median of the government subsidies divided by total revenues. Column (2) of Table 4 shows that the coefficient on the interaction term between $DM_subsidy$ and $Ccpu$ is significantly positive, suggesting that CPU has a greater inhibition effect on the labor demand of high-carbon firms that receive fewer government subsidies.

4.3. Advanced tech employee

Advanced tech employees can create higher productivity (Xie et al., 2024). Thus, we expect that firms with more advanced tech employees have lower total labor demand when firms reduce investment and scale down production due to CPU. Following Xie et al. (2024), we define 0 and 1 based on the annual city industry median for the ratio of advanced tech employees to total employees (DM_tech). Column (3) of Table 4 shows that the coefficient on the interaction term multiplier of DM_tech and $Ccpu$ is negative and significant. It is clear that CPU dampens labor demand more for high-carbon emission firms with more highly skilled employees because advanced tech employees can provide more capacity and reduce size of labor.

4.4. Profitability

Firms with weak profitability tend to invest less because of difficulties in accessing capital (García Lara et al., 2016; Liu et al., 2021), which correspondingly reduces the labor demand. As a result, firms with higher profitability have a lower negative impact of labor demand in the CPU. Following Liu et al. (2021), we employ ROE to measure profitability, and assign values of 0 and 1 according to the annual city industry median ($DM_profitability$). Column (4) of Table 4 shows that the coefficient of $DM_profitability$*$Ccpu$ is positive and significant at 1%. This suggests that the weaker the CPU reduces labor demand in highly profitable firms, supporting our hypothesis.

Table 4
Heterogeneity analysis and mechanism test results.

VARIABLES	Heterogeneity analysis				Mechanism test	
	(1) $Labor_{t+1}$	(2) $Labor_{t+1}$	(3) $Labor_{t+1}$	(4) $Labor_{t+1}$	(5) $Labor_{t+1}$	(6) $Labor_{t+1}$
Ccpu	−0.1602***	−0.1743***	−0.1826***	−0.2043***	−0.3283***	−0.1696***
	(0.0215)	(0.0264)	(0.0269)	(0.0292)	(0.0372)	(0.0277)
DM_ed	−0.2338**					
	(0.0961)					
*Ccpu*DM_ed*	0.0943***					
	(0.0357)					
DM_subsidy		−0.2033*				
		(0.1097)				
Ccpu DM_subsidy*		0.0879**				
		(0.0405)				
DM_tech			0.2927***			
			(0.0734)			
Ccpu DM_tech*			−0.0626**			
			(0.0283)			
DM_profitability				−0.2293***		
				(0.0682)		
Ccpu DM_profitability*				0.0879***		
				(0.0266)		
Giv					−0.9290***	
					(0.3091)	
*Ccpu*Giv*					0.3403**	
					(0.1343)	
Outsourcing						0.4681**
						(0.2011)
Ccpu Outsourcing*						−0.1845**
						(0.0798)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	9978	9978	9978	9978	8284	9978
R-squared	0.9295	0.9302	0.9310	0.9306	0.9469	0.9302
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Cluster	Firm	Firm	Firm	Firm	Firm	Firm

Note: *, **, *** represents significance at 10%, 5%, and 1%, respectively. Standard errors are clustered at the firm level and reported in parentheses.

5. Possible mechanisms

5.1. Green investment

Green investment is the allocation of capital aimed at reducing greenhouse gases and air pollutants (Zhang et al., 2024), and is related to the climate policies. Uncertainty increases the wait-option value and enables firms to make more conservative hiring decisions before sinking capital (e.g. green investments) (Bloom, 2009). Consequently, firms are likely to reduce labor demand until the uncertainty is resolved. Thus, in the face of CPU, firms have an incentive to reduce and delay green investments, thereby reducing the corresponding employees demand. Following Zhang et al. (2024), the green investment (Giv) is measured as the ratio of the annual green investment amount to the firms' sales. The interaction terms, $Ccpu * Giv$, are used to explore the potential channel for green investment. From Column (5) of Table 4, the positive coefficient on the interaction term confirms the above hypothesis.

5.2. Labor outsourcing

Financial friction is a crucial factor that influences firms' labor decisions. For example, Caggese et al. (2019) find that financially constrained firms tend to fire newer workers, who are associated with lower firing costs, thus there is a preference for flexible labor outsourcing (Valverde et al., 2000; Li et al., 2025). Given this, we expect that in the CPU leads to a rise in outsourcing of firms' labor (because of capital constraints), which reduces firms' labor demand. The proportion of outsourced compensation in the total employee compensation are introduced to measure labor outsourcing in this paper (Li et al., 2025),[6] represented by Outsourcing. The coefficient of $Outsourcing * Ccpu$ is shown to be negative and significant in Column (6) of Table 4. This suggests that CPU leads firms to adopt more labor outsourcing and thus has a crowding-out effect on their own labor demand. This finding further supports our baseline regression results.

[6] Outsourced labor compensation had been clarified in the notes to the financial statements before 2012 and presented independently from 2012.

6. Conclusion

This paper examines the impact of CPU on the labor demand of high-carbon firms using data from high-carbon listed companies in China from 2007 to 2022. The empirical results find that climate policy uncertainty inhibits corporate labor demand and all results remain consistent with the baseline regression results after using robustness tests. The mechanism analysis indicates that climate policy uncertainty primarily affects corporate labor demand through channels of green investment and labor outsourcing which are consistent with our theoretical description and support our hypothesis.

In addition, this paper finds some factors that can have a heterogeneous effect on the relationship between CPU and labor demand. Specifically, firms in poorer economic situation regions, firms with fewer government subsidies and poorer profitability have stronger negative association between climate policy uncertainty and corporate labor demand, mainly because they lead to capital constraints. And firms with more advanced tech employees also strengthen the CPU's inhibition to labor demand, mainly because advanced tech employees can provide more efficient production capacity.

Overall, the CPU will lead Chinese high-carbon emission firms to invest more cautiously, ultimately inhibiting their labor demand. Therefore, this paper suggests that the government may implement more effective climate policies to mitigate the adverse effects of climate policy uncertainty. In addition, by providing subsidies to alleviate the financing pressure faced by enterprises due to the transition, enterprises should be pushed to ensure sustainable development while avoiding lowering labor demand due to excessive conservatism.

CRediT authorship contribution statement

Pengju Zheng: Writing – review & editing, Writing – original draft, Supervision, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Zihao Ning:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Qingyang Wu:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization. **WeiWei Li:** Writing – review & editing, Writing – original draft, Supervision, Software, Project administration, Methodology, Investigation, Data curation, Conceptualization.

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Appendix

See Table A.1.

Table A.1
Measurements of variables.

Symbol	Measurements
Labor	Natural logarithm of the total number of employees in the enterprise.
C_{cpu}	Climate policy uncertainty index for China in a given year (Ma et al., 2023).
Age	The natural logarithm of one plus the number of years a firm has been listed.
Size	The natural logarithm of total assets at the end of the year.
Leverage	Total liability/Total asset.
Growth	The percentage change in sales from year t-1 to t.
TobinQ	The ratio of the market value of assets to the book value of assets.
Cashflow	Net cash flow from operating activities/total assets.
Board	The natural logarithm of the number of board members.
Independent	The number of independent directors divided by the total number of directors.
Dual	A dummy variable that equals 1 if the CEO is also the chairperson, and 0 otherwise.
Top1	The largest shareholder's shareholding percentage.
Unemployment	Natural logarithm of the total unemployed population in the province where the firm located in.
Loss	A dummy variable that equals 1 if net income is negative, and 0 otherwise.
IV_Cpu	Annual average of U.S. climate policy uncertainty divided by 100 (Gavriilidis, 2021).
Investment	The ratio of fixed assets to total assets (Caggese, 2007).
KZ	The degree of financing constraints (Kaplan and Zingales, 1997).
Dummy_Ed	Equals 1 if GDP per capita of a given city is greater than year median value, equals 0 otherwise.
Dummy_analysts	Equals 1 if the numbers of analysts tracking the target firm is greater than year industry median value, otherwise 0.
Epu	Annual average value of China's economic policy uncertainty index divided by 100.
Gdp_growth	The total national GDP growth divided by the GDP of the previous year.
DM_ed	Equals 1 if GDP per capita of a given city is greater than year median value, equals 0 otherwise.
DM_subsidy	Equals 1 if the government subsidy to a given firm is greater than year-city-industry median value, equals 0 otherwise.
DM_tech	Equals 1 if the ratio of advanced tech employees to total employees of a given firm is more than year-city-industry median value, equals 0 otherwise.
DM_profitability	Equals 1 if ROE of a given firm is more than year-city-industry median value, equals 0 otherwise.
Giv	The ratio of the annual green investment amount to the firms' sales.
Outsourcing	The proportion of outsourced compensation in the total employee compensation.

Data availability

Data will be made available on request.

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